

Introduction to Elementary Particles – David Griffiths

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An updated set of notes to supersede my handwritten ones in 2025.

1 A Hitchhiker's Guide to Particles

1.1 Dramatis Personae

The Standard Model includes 17 subatomic particles (but not gravity).

1. Fermions (spin 1/2)
 - 6 Leptons: $e, \mu, \tau, \nu_e, \nu_\mu, \nu_\tau$ and antileptons
 - 6 Quarks: u, d, s, c, b, t and antiquarks
2. Gauge bosons (spin 1)
 - Gluon (strong force carrier): g
 - Photon (electromagnetic force carrier): γ
 - Intermediate vector bosons (weak force carriers): W^+, W^-, Z
3. Scalar bosons (spin 0)
 - Higgs boson (gives particles mass): H

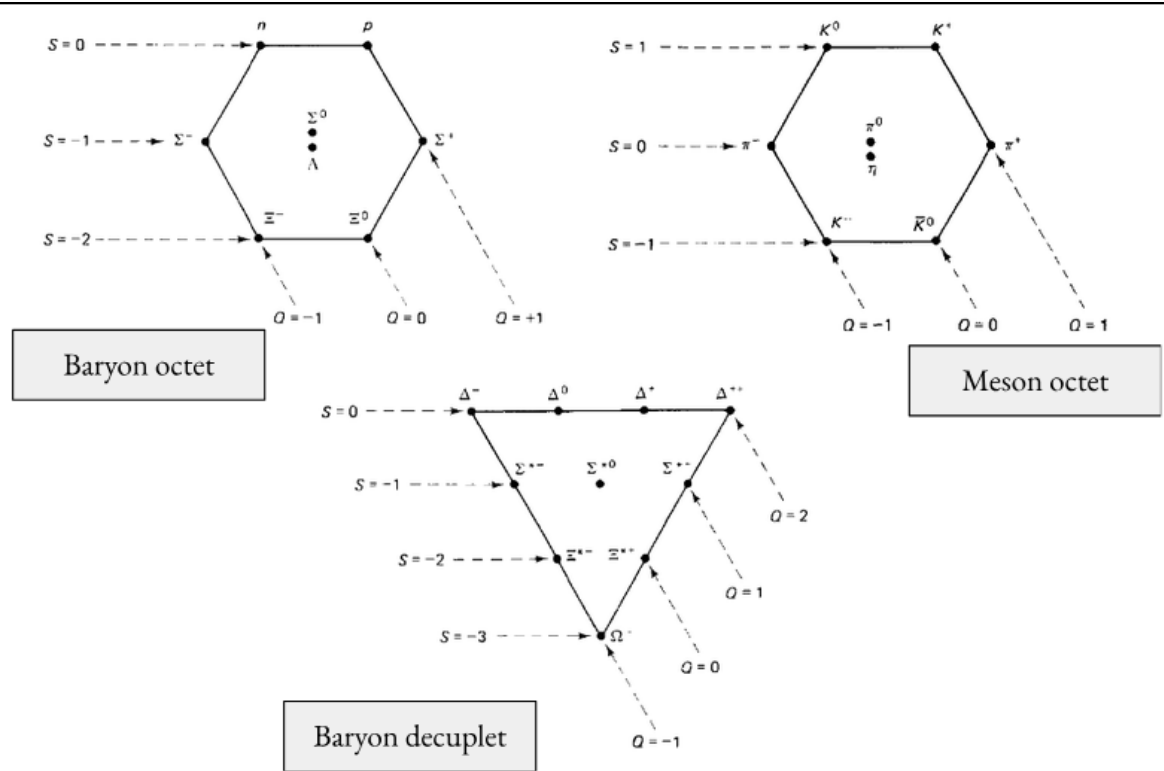
Lepton Families

	l	Q	L_e	L_μ	L_τ
1st generation	e	-1	1	0	0
	ν_e	0	1	0	0
2nd generation	μ	-1	0	1	0
	ν_μ	0	0	1	0
3rd generation	τ	-1	0	0	1
	ν_τ	0	0	0	1

Quark Families

	q	Q	D	U	S	C	B	T
1st generation	d	-1/3	-1	0	0	0	0	0
	u	2/3	0	1	0	0	0	0
2nd generation	s	-1/3	0	0	-1	0	0	0
	c	2/3	0	0	0	1	0	0
3rd generation	b	-1/3	0	0	0	0	-1	0
	t	2/3	0	0	0	0	0	1

Baryons and mesons can be organized into the baryon decuplet/octet and meson octet:



Baryon and meson octets/decuplets

1.2 Particle Dynamics

The Four Interactions

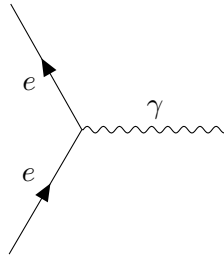
Force	Strength	Theory	Mediator
Strong	10	Chromodynamics	Gluon
Electromagnetism	10^{-2}	Electrodynamics	Photon
Weak	10^{-13}	Flavordynamics	W, Z
Gravitational	10^{-42}	Geometrodynamics	Graviton

The mediators transmit forces between 1 lepton/quark to another.

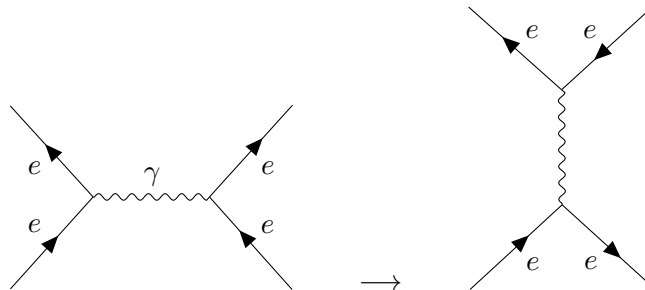
- Electromagnetism: formulated by Maxwell in 1860s.
- Quantum electrodynamics: formulated by Tomonaga, Feynman, Schwinger in 1940s.
- Weak interactions: formulated by Fermi, Lee & Yang, Feynman & Gell-Mann, etc. Until Glashow, Weinberg, Salam in 1960s.
 - GWS theory: weak & EM interactions are part of a unifying *electroweak interaction*.
- Strong interactions: formulated by Yukawa in 1930s, then others in 1970s.

1.3 Quantum Electrodynamics

All electromagnetic phenomena are reducible to the following Feynman diagram (read bottom to top):



I.e. e enters, emits/absorbs γ , and exits. The familiar Coulomb repulsion/attraction of charges (Møller scattering) is on the left. By rotating Møller scattering, we get Bhabha scattering (right): a particle-antiparticle pair annihilate, forming a photon, which later produces a particle-antiparticle pair.

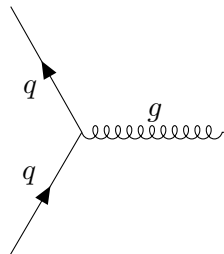


Note that only external lines describe observable particles. What goes on inside with virtual particles is allowed to violate conservation of mass, etc, as long as the external particles are all okay. Further, the lines don't accurately represent trajectories. E.g. above, the interaction might be an attraction, not repulsion. All the diagram shows is that two particles go in, interact, and two particles come out.

Each vertex introduces $\alpha = 1/137$, the coupling constant. So Feynman diagrams with many vertices, though they're physically possible, are negligible and we can sum up the possibilities.

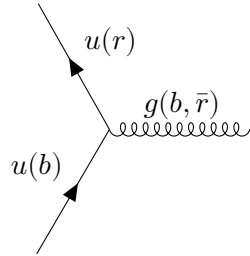
1.4 Quantum Chromodynamics

Color, not charge, causes the strong interaction. Gluons mediate the force between quarks. The fundamental vertex is

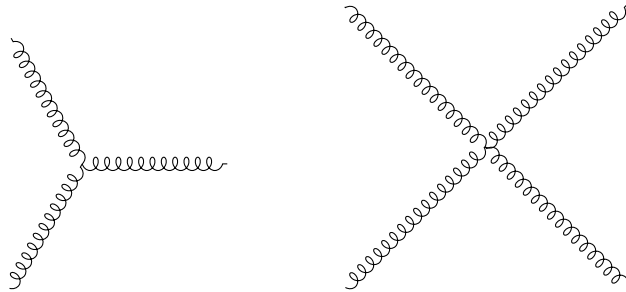


The flavor of the quark remains the same. However, color can change. E.g. blue up to red up quark,

gluon carries away difference.



Gluons themselves have color (2), so primitive gluon vertices exist (only 3-gluon and 4-gluon vertices):



The coupling constant α_S depends on the separation distance of interacting particles.

- Quark polarization drives α_S up at short distances.
- Gluon polarization drives α_S down at short distances.

$$\text{Critical parameter: } a = 2f - 11n \quad \begin{cases} a < 0 & \text{QCD coupling decreases at short dist.} \\ a > 0 & \text{QCD coupling increases at short dist.} \end{cases}$$

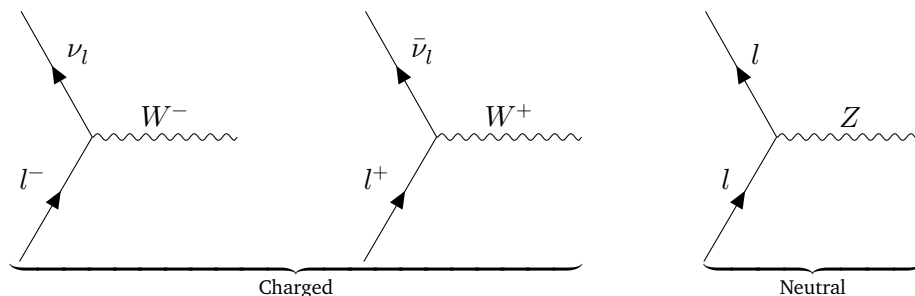
In the standard model ($f = 6$ flavors and $n = 3$ colors), $a < 0$ so QCD coupling decreases at short distances.

1.5 Weak Interactions

1.5.1 Quarks and Leptons

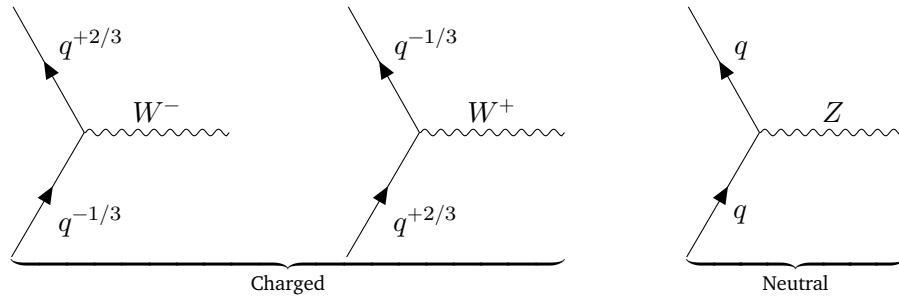
Quarks and leptons carry a “weak charge” (?) that leads to weak interactions. Weak interactions can be charged (mediated by $W^\pm$) or neutral (mediated by Z).

For leptons, the fundamental charged and neutral vertices are



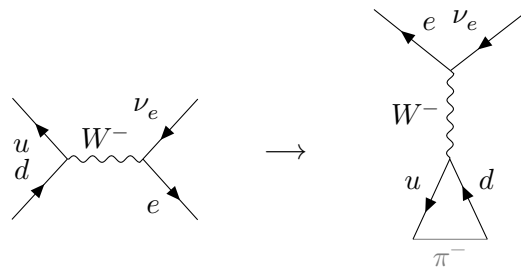
These vertices connect members to others of the **same generation**.

For quarks, same story: members of the same generation are connected (u, c, t to d, s, b , respectively). The fundamental charged and neutral vertices are

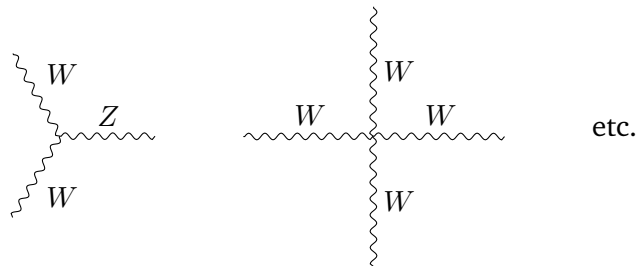


Color is conserved, but not flavor. I.e. **flavor is not conserved in weak interactions**.

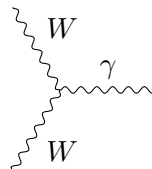
The weak interaction mediates decay. E.g. by rotating $d + \nu_e \rightarrow u + e$ (and enforcing quark confinement), we get $\pi^- \rightarrow e^- + \bar{\nu}_e$.



W and Z can also couple to each other:



Since W is charged, it can also couple to photons.



1.5.2 Skewed Quark Generations

The current available vertices don't explain strangeness-changing interactions, since the only interactions are between quarks of the same generation. Resolution: quark generations are skewed. They

couple as

$$\begin{pmatrix} u \\ d' \end{pmatrix}, \begin{pmatrix} c \\ s' \end{pmatrix}, \begin{pmatrix} t \\ b' \end{pmatrix}$$

where

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \underbrace{\begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}}_{\text{CKM matrix}} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

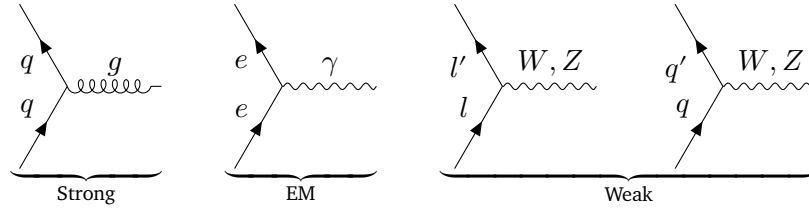
The CKM matrix is close to I , but there's some cross terms that explain strangeness-changing interactions.

1.6 Conservation Laws

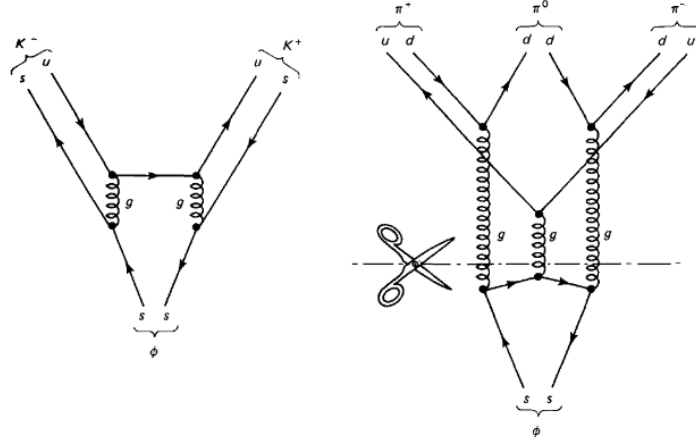
There are base kinematic conservation laws, from special relativity:

1. Conservation of energy & momentum
2. Conservation of angular momentum

There are also dynamic conservation laws for particle physics. Recall our fundamental vertices. Any quantity conserved in them must be conserved overall.



1. Conservation of charge
2. Conservation of color
3. Conservation of baryon number ($A = 1$ for baryons, $A = -1$ for antibaryons)
4. Conservation of electron number, muon number, tau number
5. Flavor is **approximately** conserved (except in weak interactions)
6. OZI rule: if a Feynman diagram can be split in two by only cutting across gluon lines, the process is suppressed.



The OZI rule explains why $\phi \rightarrow K^- + K^+$ is more common than $\phi \rightarrow \pi^- + \pi^0 + \pi^+$.

2 Special Relativity

2.1 Lorentz Transformations

The Lorentz transformations between frame S at rest and frame S' moving at v w.r.t. S :

$$\begin{aligned}x' &= \gamma(x - vt) \\y' &= y \\z' &= z \\t' &= \gamma\left(t - \frac{v}{c^2}x\right) \\ \gamma &= \frac{1}{\sqrt{1 - v^2/c^2}}\end{aligned}$$

Some consequences:

1. Relativity of simultaneity: Even if $t_A = t_B$, if $x_A \neq x_B$, then $t'_A \neq t'_B$ in general.

$$t'_A = t'_B + \frac{\gamma v}{c^2}(x_B - x_A)$$

2. Length contraction: $L' = L/\gamma$

3. Time dilation: $t = \gamma t'$

4. Velocity addition formula (u is measured in S , u' is measured in S' , and v is the relative velocity between S and S')

$$u = \frac{u' + v}{1 + u'v/c^2}$$

Define the 4-vector $x^\mu = (ct, x, y, z)$ such that the Lorentz transformation becomes

$$\begin{aligned}x^{0'} &= \gamma(cx^0 - \beta x^1) \\x^{1'} &= \gamma(x^1 - \beta cx^0) \\ \beta &= \frac{v}{c}\end{aligned}$$

More simply,

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$x^{\mu'} = \Lambda^\mu_{\nu'} x^\nu$$

Define $g = \text{diag}(1, -1, -1, -1)$. (Warning: this convention is reversed from most) Then the following quantity remains constant across coordinate changes:

$$I = x_\mu x^\mu \equiv x^\mu g_{\mu\nu} x^\nu = (\Delta x^0)^2 - (\Delta x^1)^2 - (\Delta x^2)^2 - (\Delta x^3)^2$$

$x_\mu = g_{\mu\nu} x^\nu$ is the covariant 4-vector, while x^μ is the original covariant 4-vector.

1. If $a_\mu a^\mu > 0$, then a^μ is timelike.
2. If $a_\mu a^\mu < 0$, then a^μ is spacelike.
3. If $a_\mu a^\mu = 0$, then a^μ is null.

2.2 Relativistic Dynamics

Proper time is invariant; everyone agrees on it regardless of their velocity (unlike coordinate time).

$$\begin{aligned}\text{Proper time: } d\tau &= \frac{dt}{\gamma} \\ \text{Proper velocity: } \vec{\eta} &= \frac{d\vec{x}}{d\tau} = \gamma \vec{v}\end{aligned}$$

Define the relativistic energy-momentum 4-vector as follows:

$$\begin{aligned}E &= \gamma mc^2 \\ p^0 &= \frac{E}{c} = \gamma mc \\ p^i &= m\eta^i = \gamma m\vec{v} \\ p^\mu &= \left(\frac{E}{c}, p_x, p_y, p_z \right)\end{aligned}$$

The quantity $p_\mu p^\mu = E^2/c^2 - \vec{p}^2 = m^2 c^2$ is invariant.

- Rest energy: $E_r = mc^2$

- Kinetic energy: $E_K = (\gamma - 1)mc^2$
- Total energy: $E_{\text{tot}} = \gamma mc^2$

If $m = 0$, $E = |\vec{p}|c$, but what is $\vec{p}$? The answer comes from QM: $E = hf$, so

$$|\vec{p}| = \frac{hf}{c}$$

In relativistic collisions, the energy-momentum 4-vector is conserved. Thus

$$\sum p_i^\mu = \sum p_f^\mu$$

Note that the quantity $p_\mu p^\mu$ is always invariant, leading to the following equation:

$$E^2 = \vec{p}^2 c^2 + (mc^2)^2$$

3 Symmetries

3.1 Spin and Orbital Angular Momentum

Symmetries are operations that leave the system invariant. I.e. rotation by 120° on an equilateral triangle, or multiplication by I_2 on the set of 2×2 matrices.

The set of symmetry operations forms a group:

- Closed under products
- Existence of identity and inverse
- Associative

Groups in Particle Physics	
Group Name	Matrices in Group
$U(n)$	$n \times n$ unitary group, $U^*U = I$
$SU(n)$	$n \times n$ unitary group with determinant 1
$O(n)$	$n \times n$ orthogonal group, $OO = I$
$SO(n)$	$n \times n$ orthogonal group with determinant 1

Angular momentum has 2 categories: orbital angular momentum $L = rmv$ and spin angular momentum $L = I\omega$. A measurement of just L_x , L_y , or L_z affects others. Instead, we measure $\mathbf{L}^2$ along with one component, e.g. L_z .

For orbital angular momentum,

- Measuring $\mathbf{L}^2$ yields $l(l+1)\hbar^2$, for $l = 0, 1, 2, \dots$
- Measuring L_z yields $m_l\hbar$, for $m_l = -l, -l+1, \dots, l-1, l$

For spin angular momentum,

- Measuring $\mathbf{S}^2$ yields $s(s+1)\hbar^2$, for $s = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, \dots$

- Measuring S_z yields $m_s \hbar$, for $m_s = -s, -s + 1, \dots, s - 1, s$

s is fixed for each particle, and we call it the **spin**. Fermions have half-integer spin, while bosons have integer spin.

3.1.1 Addition of Angular Momenta

To add two angular momenta, e.g. $|j_1 m_1\rangle$ and $|j_2 m_2\rangle$, where j 's are measurements of $\mathbf{J}^2$ and m 's are measurements of J_z : note that J_z adds, while $\mathbf{J}^2$ can take a range of values.

$$\begin{aligned} |j_1 m_1\rangle + |j_2 m_2\rangle &= |j m\rangle \\ j &= |j_1 - j_2|, |j_1 - j_2| + 1, \dots, |j_1 + j_2| - 1, |j_1 + j_2| \\ m &= m_1 + m_2 \end{aligned}$$

E.g. quarks and antiquarks (with spin $\frac{1}{2}$) bound with 0 angular momentum can result in a total angular momentum (j) of $\frac{1}{2} + \frac{1}{2} = 1$ or $\frac{1}{2} - \frac{1}{2} = 0$. The former are the vector mesons (ρ, K^*, ϕ, ω), while the latter are the pseudo-scalar mesons (π, K, η, η').

Clebsch-Gordon tables can help with the specific decomposition amplitudes of

$$|j_1 m_1\rangle |j_2 m_2\rangle = \sum C_{m m_1 m_2}^j |j m\rangle$$

3.1.2 Spin $\frac{1}{2}$ Notation

Define spinors:

$$|\frac{1}{2} \frac{1}{2}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\frac{1}{2} -\frac{1}{2}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The general state of a spin $\frac{1}{2}$ particle is a superposition:

$$\begin{aligned} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} &= \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \alpha &= \text{prob. that } S_z \text{ yields } \frac{1}{2}\hbar \\ \beta &= \text{prob. that } S_z \text{ yields } -\frac{1}{2}\hbar \end{aligned}$$

Also define the Pauli spin matrices:

$$\begin{aligned} \sigma_x &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ \hat{S}_i &= \frac{\hbar}{2} \sigma_i \end{aligned}$$

Eigenvalues of $\hat{S}_i$ are the possible measurement outcomes of the angular momentum.

3.2 Parity Violation

Lee, Yang, and Wu showed that parity is not conserved in weak interactions. Let the helicity of a particle be m_s/s . If $m_s = \frac{1}{2}$, the particle is “right-handed,” if $m_s = -\frac{1}{2}$, the particle is “left-handed.”

The helicity of a particle is invariant if the particle is massless.

The helicity of neutrinos is **observed** to be fixed. All neutrinos are left-handed, and all antineutrinos are right-handed. However, the neutrino may have a small mass, so its helicity might be frame-dependent.

- There’s also CP violation, where C is charge conjugation (particle to its antiparticle).
- However, it’s conjectured through QFT that TCP (where T is time reversal) is conserved.

4 Bound States

4.1 Hydrogen, e^-p^+

We start with the Schrodinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$$

By assuming a separable solution $\psi = \psi(x, y, z)T(t)$, we can get $\psi = \psi(x, y, z)e^{-iEt/\hbar}$. Assume a spherically symmetric potential $V(r)$. Separating even further as $\psi(x, y, z) = R(r)\Theta(\theta)\Phi(\phi)$. The solutions to Θ and Φ are given by

$$P_l^{m_l}(z) = \frac{1}{2^l l!} (1-z^2)^{m_l/2} \left(\frac{d}{dz} \right)^{l+m_l} (z^2-1)^l \quad (\text{Legendre function})$$

$$Y_l^{m_l}(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m_l)!}{(l+m_l)!}} (-1)^{m_l} P_l^{m_l}(\cos \theta) e^{im_l \phi} \quad (\text{Spherical harmonic})$$

Introduce $u(r) = rR(r)$ to get the radial Schrodinger equation. The potential includes a centrifugal barrier of $\mathbf{L}^2/2mr^2 = l(l+1)\hbar^2/2mr^2$.

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[V(r) + \frac{\hbar^2}{2m} \frac{l(l+1)}{r^2} \right] u = Eu$$

$$\psi = T(t)R(r)\Theta(\theta)\Phi(\phi) = e^{-iEt/\hbar} u(r) Y_l^{m_l}(\theta, \phi)$$

Hydrogen (e^-p^+) has allowed energy levels at

$$E_n = -\frac{me^4}{2\hbar^2 n^2} = -\alpha^2 \frac{mc^2}{2n^2} \quad , \quad \alpha = \frac{e^2}{\hbar c} \approx \frac{1}{137}$$

The emitted wavelength can take values of

$$E = hf = -\frac{me^4}{2\hbar^2} \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

Thus the full wavelength spectrum can be calculated, with R as the Rydberg constant. Bohr's hydrogen model successfully matched empirical observations of the previous Rydberg model.

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \quad , \quad R = \frac{mc^4}{4\pi\hbar^3}$$

4.2 Corrections to the Hydrogen Model

4.2.1 Fine Structure (Relativistic, Spin-Orbit Coupling)

From perturbation theory, given a classical Hamiltonian H and changes ΔH , energy levels change as $\Delta E_j \approx \langle \Delta H \rangle_j$. The fine structure of hydrogen is largely due to 2 factors: relativistic effects and spin-orbit coupling.

1. **Relativistic correction:** by expanding $T_{\text{rel}} = \sqrt{\mathbf{p}^2 c^2 - (mc^2)^2} - mc^2$, we get the lowest order correction to H to be

$$\Delta H_{\text{rel}} = -\frac{\mathbf{p}^4}{8m^3c^2}$$

Then by calculating expectation values,

$$\Delta E_{\text{rel}} \approx -\alpha^4 mc^2 \frac{1}{4n^2} \left[\frac{2n}{(l + \frac{1}{2})} - \frac{3}{2} \right]$$

2. **Spin-orbit coupling:** From the electron's POV, the proton rotates around it, creating a magnetic field. The proton's "current" is $I = e/(2\pi r/v)$. In Gaussian units, the magnetic field is

$$B = \frac{2\pi I}{cr} = \frac{ev}{cr^2} = \frac{e}{mcr^3} \mathbf{L}$$

The magnetic dipole is $\vec{\mu} = -(e/mc)\vec{S}$. Thus the energy of the magnetic dipole is

$$E = -\vec{\mu} \cdot \vec{B} = \frac{e^2}{m^2 c^2 r^3} (\vec{L} \cdot \vec{S})$$

There's another subtle effect: **Thomas precession**, which introduces a 1/2 factor from the Lorentz transformations to follow the motion of the electron's frame. Thus

$$\Delta H_{\text{so}} = \frac{e^2}{2m^2 c^2 r^3} (\vec{L} \cdot \vec{S})$$

After calculating expectation values,

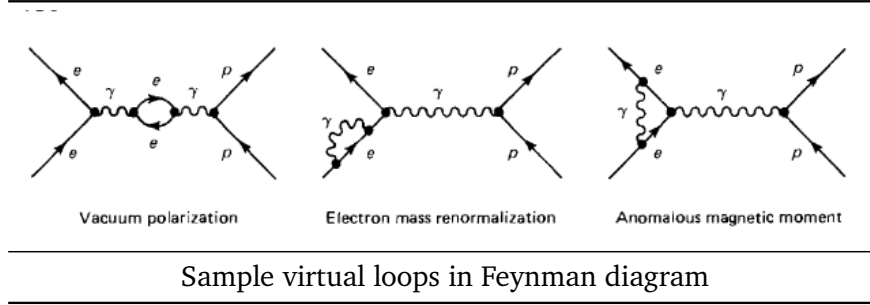
$$\Delta E_{\text{so}} = \alpha^4 mc^2 \frac{[j(j+1) - l(l+1) - \frac{3}{4}]}{4n^3 l (l + \frac{1}{2}) (l + 1)}$$

Together, the fine-structure correction is

$$E_{\text{fs}} = \alpha^4 mc^2 \frac{1}{4n^2} \left[\frac{2n}{(j + \frac{1}{2})} - \frac{3}{2} \right]$$

4.2.2 Lamb Shift

The quantization of the electromagnetic field itself leads to corrections. Vacuum loops in the Feynman diagram of the electron-proton model lead to small changes in the energy level.



$$\Delta E_{\text{Lamb}} = \begin{cases} \alpha^5 mc^2 \frac{1}{4n^3} \{k(n, 0)\} & l = 0 \\ \alpha^5 mc^2 \frac{1}{4n^3} \left\{ k(n, l) \pm \frac{1}{\pi(j+\frac{1}{2})(l+\frac{1}{2})} \right\} & l \neq 0, j = l \pm \frac{1}{2} \end{cases}$$

Where $k(n, l) \leq 0.05$.

4.2.3 Hyperfine Structure

The spin of the nucleus contributes a small correction as well. The dipole moment is similar to the one used in spin-orbit coupling, but with a γ_p factor to account for the higher angular momentum.

$$\vec{\mu}_p = \gamma_p \frac{e}{m_p c} \vec{S}_p$$

Let $\mathbf{F} = \mathbf{L} + \mathbf{S}_e + \mathbf{S}_p = \mathbf{J} + \mathbf{S}_p$ be the total angular momentum of the atom with quantum numbers f and m_f . Then the correction is

$$\Delta E_{\text{hf}} = \left(\frac{m}{m_p} \right) \alpha^4 mc^2 \frac{\gamma_p}{2n^3} \frac{\pm 1}{(f + \frac{1}{2})(l + \frac{1}{2})}, \quad f = j \pm \frac{1}{2}$$

5 The Feynman Calculus

5.1 Lifetime and Decay Rates

We're mainly interested in the lifetime of a particle and its decay rate.

τ = Mean lifetime of particle at rest

Γ = Decay rate, prob. per unit time that particle will disintegrate

Starting with $N(t)$ particles at time t , $\Gamma N(t)dt$ decay in the next dt seconds, thus

$$\begin{aligned} dN &= -\Gamma N(t)dt \\ N(t) &= N(0)e^{-\Gamma t} \end{aligned}$$

We also have the following relation between τ and Γ :

$$\tau = \frac{1}{\Gamma}$$

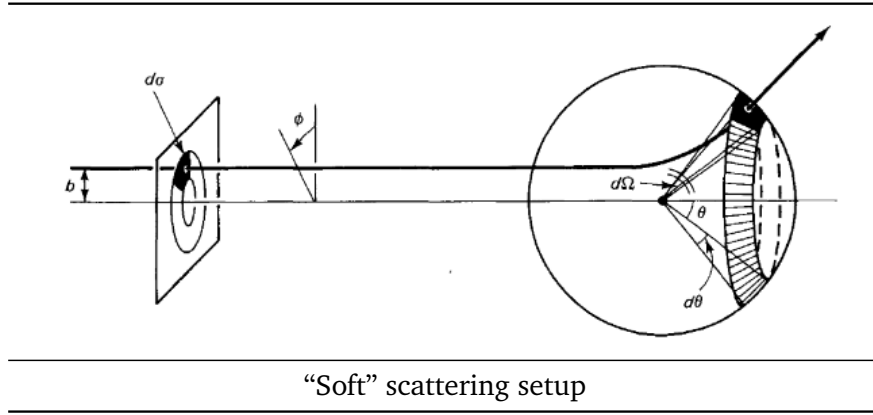
Particles can decay by different routes, e.g. a charged pion can go $\pi^+ \rightarrow \mu^+ + \nu_\mu$, $\pi^+ \rightarrow e^+ + \nu_e$, or $\pi^+ \rightarrow \mu^+ + \nu_\mu + \gamma$, etc. The total decay rate is the sum of individual decay rate paths, and we can define **branching ratios**.

$$\Gamma_{\text{tot}} = \sum_{i=1}^n \Gamma_i$$

$$\tau = \frac{1}{\Gamma_{\text{tot}}}$$

$$\text{Branching ratio} = \Gamma_i / \Gamma_{\text{tot}}$$

5.2 Scattering Particles



A particle coming from a beam area of $d\sigma$ scatters through an angle $d\Omega$. Let $D = d\sigma/d\Omega$ be the **differential cross-section**. We have

$$d\sigma = |b db d\phi|, \quad d\Omega = |\sin \theta d\theta d\phi|$$

$$D(\theta) = \frac{d\sigma}{d\Omega} = \left| \frac{b}{\sin \theta} \left(\frac{db}{d\theta} \right) \right|$$

The total cross-section is given by

$$\sigma = \int d\sigma = \int D(\theta) d\Omega = \int |D(\theta) \sin \theta| d\theta d\phi$$

For example, a “hard scattering” sphere with actual collisions would have

$$b = R \cos(\theta/2)$$

$$D(\theta) = \frac{R^2}{4}$$

$$\sigma = \int \frac{R^2}{4} d\Omega = \pi R^2$$

We can also imagine a beam of incoming particles with uniform luminosity $\mathcal{L}$ (particles per unit time per unit area). Then $dN = \mathcal{L}d\sigma$ particles pass through $d\sigma$ per unit time. “The event rate is the cross section times the luminosity.”

$$dN = \mathcal{L}d\sigma = \mathcal{L}D(\theta)d\Omega$$

$$D(\theta) = \frac{d\sigma}{d\Omega} = \frac{1}{\mathcal{L}} \frac{dN}{d\Omega}$$

5.3 Fermi’s Golden Rule

To calculate decay rates and scattering cross sections, we need the amplitude of the process ($\mathcal{M}$) and the available phase space. **Fermi’s golden rule** calculates the transition rate for a process:

$$\text{Transition rate} = \frac{2\pi}{\hbar} |\mathcal{M}|^2 \times (\text{phase space})$$

5.3.1 Golden Rule for Decays

If particle 1 decays as $1 \rightarrow 2 + 3 + \dots + n$ and particle 1 is at rest, the decay rate is

$$d\Gamma = |\mathcal{M}|^2 \frac{S}{2\hbar m_1} \left[\left(\frac{cd^3\mathbf{p}_2}{(2\pi)^3 2E_2} \right) \left(\frac{cd^3\mathbf{p}_3}{(2\pi)^3 2E_3} \right) \dots \left(\frac{cd^3\mathbf{p}_n}{(2\pi)^3 2E_n} \right) \right] \times$$

$$(2\pi)^4 \delta^4(p_1 - p_2 - \dots - p_n)$$

S is a statistical factor, contributing $1/j!$ for each group of j identical particles in the final state.

For a particle of mass m decaying into 2 massless particles,

$$\Gamma = \frac{S}{16\pi\hbar m} |\mathcal{M}|^2$$

For a particle of mass m decaying into 2 massive particles,

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c} |\mathcal{M}|^2$$

Evaluate $|\mathcal{M}|$ at the momenta required by conservation of energy and momentum. $\mathbf{p}$ is the momenta of either outgoing particle.

5.3.2 Golden Rule for Scattering

If particles 1 and 2 collide as $1 + 2 \rightarrow 3 + 4 + \dots + n$, the cross section is given by

$$|\mathcal{M}|^2 \frac{\hbar S}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{cd^3\mathbf{p}_3}{(2\pi)^3 2E_3} \right) \left(\frac{cd^3\mathbf{p}_4}{(2\pi)^3 2E_4} \right) \dots \left(\frac{cd^3\mathbf{p}_n}{(2\pi)^3 2E_n} \right) \right] \times$$

$$(2\pi)^4 \delta^4(p_1 + p_2 - p_3 - p_4 - \dots - p_n)$$

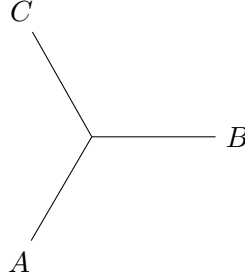
For a 2-body scattering $1 + 2 \rightarrow 3 + 4$ in the CM frame of 1 and 2, where $\mathbf{p}_i$ is the momentum of any incoming particle and $\mathbf{p}_f$ is the momentum of any outgoing particle,

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi} \right)^2 \frac{S|\mathcal{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

5.4 Feynman Rules for a Toy Theory

5.4.1 Process

Now we seek to determine $\mathcal{M}$. A toy theory: imagine a world with 3 particles A, B, C . All have spin 0 and are their own antiparticles. A can decay into B and C . The fundamental vertex is:



To determine $\mathcal{M}$:

1. Label incoming & outgoing 4-momenta with $p_1, p_2, \dots, p_n$ and internal 4-momenta lines with $q_1, q_2, \dots, q_n$. Direction is arbitrary for internal lines.
2. For each vertex, write $-ig$.
3. For each internal line, write

$$\frac{i}{q_j^2 - m_j^2 c^2}$$

4. For each vertex, write (k positive if going into vertex, k negative if going out)

$$(2\pi)^4 \delta^4(k_1 + k_2 + k_3)$$

5. For each internal line, write

$$\frac{1}{(2\pi)^4} d^4 q_j$$

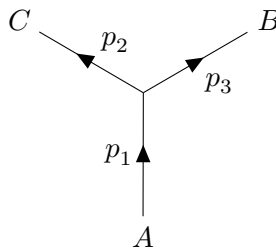
6. Integrate over all internal momenta q_j . The result involves a delta function:

$$(2\pi)^4 \delta^4(p_1 + p_2 + \dots - p_n)$$

Cancel this to enforce energy & momentum conservation. The result is $-i\mathcal{M}$.

5.4.2 Simple Examples

For the following Feynman diagram:

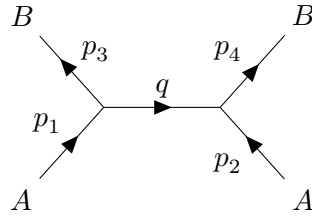


Follow the process: 1) Labels, 2) Pick up $-ig$ at vertex, 3) N/A, 4) Pick up $(2\pi)^4\delta^4(p_1 - p_2 - p_3)$, 5) N/A, 6) Erase delta function and get $-i\mathcal{M} = -ig$. Thus $\mathcal{M} = g$ for this diagram.

$$\begin{aligned}\text{Decay rate: } \Gamma &= \frac{g^2|\mathbf{p}|}{8\pi\hbar m_A^2 c} \\ \text{Mean lifetime: } \tau &= \frac{8\pi\hbar m_A^2 c}{g^2|\mathbf{p}|}\end{aligned}$$

here $\mathbf{p}$ is the magnitude of either outgoing momentum.

Or consider the example of scattering:



Following the rules: 1) Labels, 2) Two vertices contribute $(-ig)^2$, 3) Internal propagator contributes $i/(q^2 - m_C^2 c^2)$, 4) Vertices contribute $(2\pi)^4\delta^4(p_1 - p_3 - q)$ and $(2\pi)^4\delta^4(p_2 + q - p_4)$, 5) Propagator contributes $d^4q/(2\pi)^4$, 6) Combining them all gives

$$\begin{aligned}\text{Result} &= -ig^2(2\pi)^4 \int \frac{1}{q^2 - m_C^2 c^2} \delta^4(p_1 - p_3 - q) \delta^4(p_2 + q - p_4) d^4q \\ &= \frac{-ig^2}{(p_4 - p_2)^2 - m_C^2 c^2} (2\pi)^4 \delta^4(p_1 + p_2 - p_3 - p_4)\end{aligned}$$

Thus the amplitude is

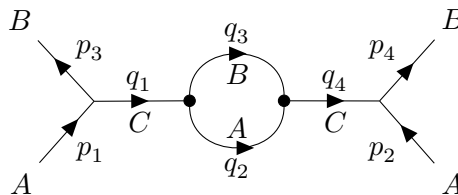
$$\mathcal{M} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2}$$

But we could also “twist” the B lines to interchange p_3 and p_4 . So overall, the amplitude to order g^2 is

$$\mathcal{M} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2} + \frac{g^2}{(p_3 - p_2)^2 - m_C^2 c^2}$$

5.4.3 Renormalization

Consider the following diagram:



The amplitude of the process turns out to be

$$\mathcal{M} = i \left(\frac{g}{2\pi} \right)^4 \frac{1}{[(p_1 - p_3)^2 - m_C^2 c^2]^2} \int \frac{1}{[(p_1 - p_3 - q)^2 - m_A^2 c^2] (q^2 - m_B^2 c^2)} d^4 q$$

After replacing $d^4 q = q^3 dq d\Omega$, at high q the integral goes to $\int_0^\infty \frac{1}{q^4} q^3 \rightarrow \infty$. To “sweep the infinities under the rug,” **regularize** the integral by introducing a cutoff factor:

$$-\frac{M^2 c^2}{(q^2 - M^2 c^2)}, \quad M \gg 1$$

The resulting integral can be separated into a finite term and a term containing M that blows up as $M \rightarrow \infty$. All M -dependent terms appear as additions to masses (m) and coupling constants (g). Thus the physical masses and couplings are interpreted to be

$$\begin{aligned} m_{\text{physical}} &= m + \delta m \\ g_{\text{physical}} &= g + \delta g \\ \delta m, \delta g &\rightarrow \infty \quad \text{as} \quad M \rightarrow \infty \end{aligned}$$

Account for infinities using the physical values of m, g in Feynman rules. Then disregard the divergent terms.

6 Quantum Electrodynamics

6.1 The Dirac Equation

Start from energy conservation $E^2 - \mathbf{p}^2 c^2 = (mc^2)^2$, and rewrite $p^\mu p_\mu - m^2 c^2 = 0$. The momentum behaves just like a quantum operator, $p_\mu \rightarrow i\hbar \partial_\mu$ in this sign convention. Thus we get

$$\begin{aligned} -\hbar^2 \partial^\mu \partial_\mu \psi - m^2 c^2 \psi &= 0 \\ -\frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} + \nabla^2 \psi &= \left(\frac{mc}{\hbar} \right)^2 \psi \end{aligned}$$

This is the **Klein-Gordon equation**. Similarly, the Dirac equation is obtained from the energy conservation relation. We factor as

$$\begin{aligned} p^\mu p_\mu - m^2 c^2 &= (\gamma^\kappa p_\kappa + mc)(\gamma^\lambda p_\lambda - mc) = 0 \\ \gamma^0 &= \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \\ \sigma_i &= \text{Pauli spin matrices} \end{aligned}$$

The conventional choice is $\gamma^\kappa p_\kappa - mc = 0$. Substitute $p_\mu \rightarrow i\hbar \partial_\mu$ to get the **Dirac equation**:

$$i\hbar \gamma^\mu \partial_\mu \psi - mc\psi = 0$$

6.2 Solutions to the Dirac Equation

6.2.1 Stationary Particles

Assume ψ is independent of position, i.e. $\partial_i \psi = 0$. Let $\psi = (\psi_A \ \psi_B)$, where $\psi_A = (\psi_1 \ \psi_2)$ and $\psi_B = (\psi_3 \ \psi_4)$. Solving the Dirac equation gives

$$\begin{cases} \frac{\partial \psi_A}{\partial t} = -i \frac{mc^2}{\hbar} \psi_A \longrightarrow \psi_A(t) = e^{-i(mc^2/\hbar)t} \psi_A(0) \\ -\frac{\partial \psi_B}{\partial t} = -i \frac{mc^2}{\hbar} \psi_B \longrightarrow \psi_B(t) = e^{+i(mc^2/\hbar)t} \psi_B(0) \end{cases}$$

From quantum mechanics, we know that a quantum state with time-dependence evolves with a factor $e^{-iEt/\hbar}$. This is represented here with $E = mc^2$ (rest energy). But ψ_B seems to have $E = -mc^2$. We interpret this negative energy as an **antiparticle** with energy E .

Overall there are 4 solutions:

$$\begin{aligned} \psi^{(1)} &= e^{-i(mc^2/\hbar)t} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, & \psi^{(2)} &= e^{-i(mc^2/\hbar)t} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \\ \psi^{(3)} &= e^{+i(mc^2/\hbar)t} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, & \psi^{(4)} &= e^{+i(mc^2/\hbar)t} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$$

$\psi^{(1)}$ and $\psi^{(2)}$ correspond to a particle with spin up/down, while $\psi^{(3)}$ and $\psi^{(4)}$ correspond to an antiparticle with spin up/down.

6.2.2 Plane Waves

We assume a plane-wave solution $\psi(x^\mu) = ae^{-i(i/\hbar)x^\mu p_\mu} u(p^\mu)$. The Dirac equation becomes

$$\begin{aligned} \gamma^\mu p_\mu a e^{-(i/\hbar)x^\mu p_\mu} u - mca e^{-(i/\hbar)x^\mu p_\mu} u &= 0 \\ (\gamma^\mu p_\mu - mc)u &= 0 \\ \begin{pmatrix} E/c - mc & -\mathbf{p} \cdot \boldsymbol{\sigma} \\ \mathbf{p} \cdot \boldsymbol{\sigma} & -E/c - mc \end{pmatrix} \begin{pmatrix} u_A \\ u_B \end{pmatrix} &= 0 \end{aligned}$$

The solutions (after calculations) are

$$\begin{aligned}
 u^{(1)} &= N \begin{pmatrix} 1 \\ 0 \\ \frac{c(p_z)}{E + mc^2} \\ \frac{c(p_x + ip_y)}{E + mc^2} \end{pmatrix}, & u^{(2)} &= N \begin{pmatrix} 0 \\ 1 \\ \frac{c(p_x - ip_y)}{E + mc^2} \\ \frac{c(-p_z)}{E + mc^2} \end{pmatrix} \\
 \nu^{(1)} &= N \begin{pmatrix} \frac{c(p_x - ip_y)}{E + mc^2} \\ \frac{c(-p_z)}{E + mc^2} \\ 0 \\ 1 \end{pmatrix}, & \nu^{(2)} &= -N \begin{pmatrix} \frac{c(p_z)}{E + mc^2} \\ \frac{c(p_x + ip_y)}{E + mc^2} \\ 1 \\ 0 \end{pmatrix}
 \end{aligned}$$

Particles are labeled with u and antiparticles with ν . Antiparticles are interpreted with their own energy (not the “negative energy” solution), as $\nu^{(1)} = u^{(4)}(-E, -\mathbf{p})$ and $\nu^{(2)} = -u^{(3)}(-E, -\mathbf{p})$.

6.2.3 Bilinear Covariants

The Dirac spinor transforms as (going to a frame moving at v)

$$\begin{aligned}
 \psi &\rightarrow S\psi \\
 S &= a_+ + a_- \gamma^0 \gamma^1 = \begin{pmatrix} a_+ & a_- \sigma_1 \\ a_- \sigma_1 & a_+ \end{pmatrix} \\
 a_{\pm} &= \pm \sqrt{\frac{1}{2}(\gamma - 1)}
 \end{aligned}$$

Let the **adjoint spinor** be $\bar{\psi} = \psi^\dagger \gamma^0 = (\psi_1^* \ \psi_2^* \ -\psi_3^* \ -\psi_4^*)$. The quantity $\bar{\psi}\psi = |\psi_1|^2 + |\psi_2|^2 - |\psi_3|^2 - |\psi_4|^2$ is Lorentz invariant.

- $\bar{\psi}\psi = \text{scalar}$
- $\bar{\psi}\gamma^5\psi = \text{pseudoscalar}$
- $\bar{\psi}\gamma^\mu\psi = \text{vector}$
- $\bar{\psi}\gamma^\mu\gamma^5\psi = \text{pseudovector}$
- $\bar{\psi}\sigma^{\mu\nu}\psi = \text{antisymmetric tensor}$

$$\begin{aligned}
 \sigma^{\mu\nu} &= \frac{i}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) \\
 \gamma^5 &= i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}
 \end{aligned}$$

6.3 The Photon

Using tensors, Maxwell's equations become

$$\begin{cases} \nabla \cdot \mathbf{E} = 4\pi\rho \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} = \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} + \frac{4\pi}{c} \mathbf{J} \end{cases} \longrightarrow \begin{cases} \partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu \\ \partial_\mu \partial^\mu A^\nu - \partial^\nu (\partial_\mu A^\mu) = \frac{4\pi}{c} J^\nu \end{cases}$$

The EM field strength tensor, current 4-vector, and potential are defined as

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}$$

$$J^\mu = (c\rho, \mathbf{J})$$

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad A^\mu = (V, \mathbf{A})$$

Because from Maxwell's equation,

$$\mathbf{B} = \nabla \times \mathbf{A}$$

$$\nabla \times \left(\mathbf{E} + \frac{1}{c} \frac{\partial \mathbf{A}}{\partial t} \right) = 0 \longrightarrow \mathbf{E} = -\nabla V - \frac{1}{c} \frac{\partial \mathbf{A}}{\partial t}$$

This can be written as

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

A transformation $A'_\mu \rightarrow A_\mu + \partial_\mu \lambda$ leaves $F^{\mu\nu}$ unchanged. Such a transformation of the potential that leaves the field invariant is called a **gauge transformation**.

In light of gauge transformations, we can introduce:

1. Lorentz condition: $\partial_\mu A^\mu = 0$. Maxwell's equations become

$$\square A^\mu = \frac{4\pi}{c} J^\mu, \quad \square = \partial^\mu \partial_\mu$$

2. Coulomb gauge: $\nabla \cdot \mathbf{A} = 0$ by requiring $A^\mu = 0$ in empty space ($J^\mu = 0$).

In QED, A^μ is interpreted as the wavefunction of the photon. In free space ($J^\mu = 0$), the Lorentz condition implies the Klein-Gordon equation for $m = 0$.

$$\square A^\mu = 0$$

For plane-wave solutions $A^\mu(x) = a e^{-(i/\hbar)p \cdot x} \epsilon^\mu(p)$,

1. Lorentz gauge: $p^\mu \epsilon_\mu = 0$
2. Coulomb gauge: $\epsilon^0 = 0, \vec{\epsilon} \cdot \vec{p} = 0$

6.4 Feynman Rules for QED

6.4.1 Recap: Electrons, Protons, & Photons

Electrons (u)

$$\begin{aligned}
 \text{Plane wave: } & \psi(x) = ae^{-(i/\hbar)p \cdot x} u^{(s)}(p), \quad s = 1, 2 \\
 \text{Dirac equation: } & (\gamma^\mu p_\mu - mc)u = 0 \\
 & \bar{u}(\gamma^\mu p_\mu - mc) = 0, \quad \bar{u} = u^\dagger \gamma^0 \\
 \text{Normalized: } & \bar{u}u = 2mc \\
 \text{Orthogonal: } & \bar{u}^{(1)}u^{(2)} = 0 \\
 \text{Complete: } & \sum_{s=1,2} u^{(s)}\bar{u}^{(s)} = \gamma^\mu p_\mu + mc
 \end{aligned}$$

Protons (ν)

$$\begin{aligned}
 \text{Plane wave: } & \psi(x) = ae^{-(i/\hbar)p \cdot x} \nu^{(s)}(p), \quad s = 1, 2 \\
 \text{Dirac equation: } & (\gamma^\mu p_\mu + mc)\nu = 0 \\
 & \bar{\nu}(\gamma^\mu p_\mu + mc) = 0, \quad \bar{\nu} = \nu^\dagger \gamma^0 \\
 \text{Normalized: } & \bar{\nu}\nu = -2mc \\
 \text{Orthogonal: } & \bar{\nu}^{(1)}\nu^{(2)} = 0 \\
 \text{Complete: } & \sum_{s=1,2} \nu^{(s)}\bar{\nu}^{(s)} = \gamma^\mu p_\mu - mc
 \end{aligned}$$

Photons (ϵ)

$$\begin{aligned}
 \text{Plane wave: } & A^\mu(x) = ae^{-(i/\hbar)p \cdot x} \epsilon_{(s)}^\mu, \quad s = 1, 2 \\
 \text{Lorentz condition: } & \epsilon^\mu p_\mu = 0 \\
 \text{Coulomb gauge: } & \epsilon^0 = 0, \quad \vec{\epsilon} \cdot \vec{p} = 0 \\
 \text{Normalized: } & \epsilon^{\mu*} \epsilon_\mu = 1 \\
 \text{Orthogonal: } & \bar{\epsilon}_{(1)}^{\mu*} \epsilon_{(2)\mu} = 0 \\
 \text{Complete: } & \sum_{s=1,2} (\epsilon_{(s)})_i (\epsilon_{(s)}^*)_j = \delta_{ij} - \hat{p}_i \hat{p}_j
 \end{aligned}$$

6.4.2 Feynman Rules

From each Feynman diagram up to the desired order, follow the procedure:

1. Label incoming/outgoing momenta $p_1, p_2, \dots, p_n$ and corresponding spins $s_1, s_2, \dots, s_n$. Internal lines are $q_1, q_2, \dots, q_n$. Arrows on external lines are assigned to distinguish electrons/positrons, etc. Arrows on internal lines are assigned such that every vertex has an incoming arrow and an outgoing arrow. Arrows of external photons are forward in time.

2. External lines contribute the following factors:

Electron, incoming:	u
Electron, outgoing:	$\bar{u}$
Positron, incoming:	$\bar{\nu}$
Positron, outgoing:	ν
Photon, incoming:	ϵ^μ
Photon, outgoing:	$\epsilon^{\mu*}$

3. Each vertex contributes $ig_e\gamma^\mu$, where $g_e = \sqrt{4\pi\alpha}$.

4. Each internal line contributes a factor:

$$\begin{aligned} \text{Electrons, Positrons: } & \frac{i(\gamma^\mu q_\mu + mc)}{q^2 - m^2c^2} \\ \text{Photons: } & \frac{-ig_{\mu\nu}}{q^2} \end{aligned}$$

5. Each vertex contributes (k 's positive if incoming, negative if outgoing)

$$(2\pi)^4 \delta^4(k_1 + k_2 + k_3)$$

6. Each internal line contributes

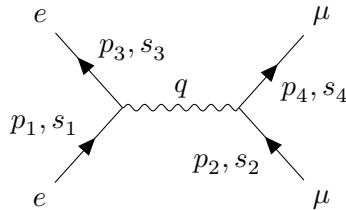
$$\frac{d^4q}{(2\pi)^4}$$

7. Integrate and cancel a delta function $(2\pi)^4 \delta^4(p_1 + p_2 + \dots - p_n)$. The result is $-i\mathcal{M}$.

8. Include a minus sign between diagrams that differ only in the exchange of two incoming/outgoing electrons or positrons.

6.4.3 Examples

For electron-muon scattering:



Following the Feynman rules, we get

$$\begin{aligned} -i\mathcal{M} &= [\bar{u}^{(s_3)}(p_3)\gamma^\mu u^{(s_1)}(p_1)] [\bar{u}^{(s_4)}(p_4)\gamma_\mu u^{(s_2)}(p_2)] \cdot \frac{ig_e^2}{(p_1 - p_3)^2} \\ \mathcal{M} &= -\frac{g_e^2}{(p_1 - p_3)^2} [\bar{u}^{(s_3)}(p_3)\gamma^\mu u^{(s_1)}(p_1)] [\bar{u}^{(s_4)}(p_4)\gamma_\mu u^{(s_2)}(p_2)] \end{aligned}$$

Electron-electron scattering is similar, but we can “cross” one outgoing momentum over the other to create a new diagram. E.g. swap (p_3, s_3) with (p_4, s_4) . Then the Feynman diagrams differ only by an exchange of fermions, so there’s a relative minus sign:

$$\mathcal{M} = -\frac{g_e^2}{(p_1 - p_3)^2} [\bar{u}(p_3)\gamma^\mu u(p_1)] [\bar{u}(p_4)\gamma_\mu u(p_2)] \\ + \frac{g_e^2}{(p_1 - p_4)^2} [\bar{u}(p_4)\gamma^\mu u(p_1)] [\bar{u}(p_3)\gamma_\mu u(p_2)]$$

6.5 Casimir’s Trick

Often we don’t know our initial spin states. We just have an ensemble of varying spin states and need to average over the initial spins, then sum over the final spins. **Casimir’s trick** reduces the calculation of $\langle |\mathcal{M}|^2 \rangle$ to an evaluation of the trace of a matrix.

$$\sum_{\text{all spin}} [\bar{u}(a)\Gamma_1 u(b)] [\bar{u}(a)\Gamma_2 u(b)]^* = \text{Tr} [\Gamma_1(\not{p}_b + m_b c)\bar{\Gamma}_2(\not{p}_a + m_a c)]$$

Here $\not{p} = p^\mu \gamma_\mu$ and $\bar{\Gamma} = \gamma_0 \Gamma^\dagger \gamma_0$. Various trace theorems then help simplify the result further (see Griffiths pp. 238–239)

For example, in electron-muon scattering, we had

$$\mathcal{M} = -\frac{g_e^2}{(p_1 - p_3)^2} [\bar{u}^{(s_3)}(p_3)\gamma^\mu u^{(s_1)}(p_1)] [\bar{u}^{(s_4)}(p_4)\gamma_\mu u^{(s_2)}(p_2)] \\ |\mathcal{M}|^2 = \frac{g_e^4}{(p_1 - p_3)^4} [\bar{u}(3)\gamma^\mu u(1)] [\bar{u}(4)\gamma_\mu u(2)] [\bar{u}(3)\gamma^\mu u(1)]^* [\bar{u}(4)\gamma_\mu u(2)]^*$$

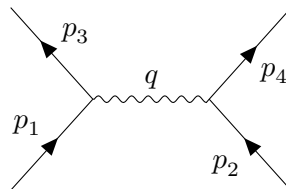
Applying Casimir’s trick (with m as the mass of the electron and M as the mass of the muon), then averaging with a factor of $\frac{1}{4}$ for 2 particles and 2 spin orientations,

$$\langle |\mathcal{M}|^2 \rangle = \frac{g_e^4}{4(p_1 - p_3)^4} \text{Tr} [\gamma^\mu(\not{p}_1 + mc)\gamma^\nu(\not{p}_3 + mc)] \times \text{Tr} [\gamma_\mu(\not{p}_2 + Mc)\gamma_\nu(\not{p}_4 + Mc)]$$

7 Electrodynamics of Quarks and Hadrons

7.1 The Rosenbluth Formula

Consider electron-proton scattering. Now we know the proton isn’t a point mass (it’s made of quarks). Thus blindly applying Feynman rules won’t work anymore.



From Casimir's trick,

$$\langle |\mathcal{M}|^2 \rangle = \frac{g_e^4}{q^4} L_{\text{electron}}^{\mu\nu} K_{\mu\nu \text{proton}}$$

$$L_{\text{electron}}^{\mu\nu} = 2\{p_1^\mu p_3^\nu + p_1^\nu p_3^\mu + g^{\mu\nu} [(mc)^2 - (p_1 \cdot p_3)]\}$$

Where the specific structure of $K_{\mu\nu}$ is uncertain. To probe $K_{\mu\nu}$: it must depend on q , $p_2 \equiv p$, and $p_4 = q + p_2$. $L^{\mu\nu}$ is symmetric, so the antisymmetric part of $K_{\mu\nu}$ contributes nothing, and the general tensor is

$$K_{\text{proton}}^{\mu\nu} = -K_1 g^{\mu\nu} + \frac{K_2}{(Mc)^2} p^\mu p^\nu + \frac{K_4}{(Mc)^2} q^\mu q^\nu + \frac{K_5}{(Mc)^2} (p^\mu q^\nu + p^\nu q^\mu)$$

$$K_4 = \frac{(Mc)^2}{q^2} K_1 + \frac{1}{4} K_2, \quad K_5 = \frac{1}{2} K_2$$

$$\rightarrow K_{\text{proton}}^{\mu\nu} = K_1 \left(-g^{\mu\nu} + \frac{q^\mu q^\nu}{q^2} \right) + \frac{K_2}{(Mc)^2} \left(p^\mu + \frac{1}{2} q^\mu \right) \left(p^\nu + \frac{1}{2} q^\nu \right)$$

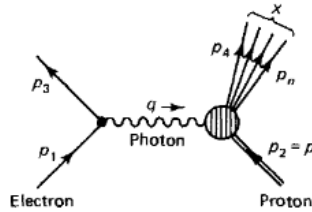
Work in the proton's frame, so $p = (Mc, 0, 0, 0)$, $p_1 = \frac{E}{c}(1, \hat{p}_i)$, and $p_3 = \frac{E'}{c}(1, \hat{p}_f)$. Here $\hat{p}_i \cdot \hat{p}_f = \cos \theta$, where θ is the scattering angle. Assume the electron is energetic enough such that $E, E' \gg mc^2$, and m can be ignored in the equation. Heavy calculation and simplification gives

$$\langle |\mathcal{M}|^2 \rangle = \frac{g_e^4 c^2}{4EE' \sin^4(\theta/2)} \left(2K_1 \sin^2 \frac{\theta}{2} + K_2 \cos^2 \frac{\theta}{2} \right)$$

$$E' = \frac{E}{1 + (2E/Mc^2) \sin^2(\theta/2)}$$

7.2 Inelastic Electron-Proton Scattering

The Rosenbluth formula describes elastic electron-proton scattering. But the scattering might be inelastic: with enough energy, all sorts of things *inside* the proton might come out and form intermediaries.



Inelastic scattering example (quarks come out)

The transition amplitude with Casimir's trick is

$$\langle |\mathcal{M}|^2 \rangle = \frac{g_e^4}{q^4} L_{\text{electron}}^{\mu\nu} K_{\mu\nu}$$

Here $K_{\mu\nu}$ describes the process $\gamma + p_2 \rightarrow p_4 + \dots + p_n$.

Fermi's golden rule for scattering gives

$$d\sigma = \frac{\hbar^2 \langle |\mathcal{M}|^2 \rangle}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left[\left(\frac{cd^3 \mathbf{p}_3}{(2\pi)^3 2E_3} \right) \cdots \left(\frac{cd^3 \mathbf{p}_n}{(2\pi)^3 2E_n} \right) \right] \times (2\pi)^4 \delta^4(p_1 + p_2 - p_3 - \cdots - p_n)$$

We find the “inclusive” cross section that includes all accessible final states and all possible outgoing momenta.

$$d\sigma = \frac{\hbar^2 g_e^4 L^{\mu\nu}}{4q^4 \sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}} \left(\frac{cd^3 \mathbf{p}_3}{(2\pi)^3 2E_3} \right) 4\pi M W_{\mu\nu}$$

$$W_{\mu\nu} = \frac{1}{4\pi M} \sum_{\text{outgoing}} \int \cdots \int K_{\mu\nu} \left(\frac{cd^3 \mathbf{p}_4}{(2\pi)^3 2E_4} \right) \cdots \left(\frac{cd^3 \mathbf{p}_n}{(2\pi)^3 2E_n} \right) \times (2\pi)^4 \delta^4(q + p - p_4 - \cdots - p_n)$$

By simplifying and setting $E, E' \gg mc^2$ and thus $m \rightarrow 0$, we get

$$\frac{d\sigma}{dE' d\Omega} = \left(\frac{\alpha \hbar}{2E \sin^2(\theta/2)} \right)^2 [2W_1 \sin^2(\theta/2) + W_2 \cos^2(\theta/2)]$$

We can obtain the Rosenbluth formula (elastic scattering) with

$$W_{1,2}(q^2, x) = -\frac{K_{1,2}(q^2)}{2Mq^2} \delta(x - 1)$$

$$x \equiv -\frac{q^2}{2q \cdot p}$$

A naive model treats a proton as a point charge, i.e.

$$K_1 = -q^2, \quad K_2 = (2Mc)^2$$

7.3 The Quark Model

At high energies, the virtual photon interacts with a single, almost-free quark. The structure for scattering off a point-mass quark of flavor i .

$$W_1^i = \frac{Q_i^2}{2m_i} \delta(x_i - 1), \quad W_2^i = -\frac{2m_i c^2 Q_i^2}{q^2} \delta(x_i - 1)$$

$$x \equiv -\frac{q^2}{2q \cdot p_i}$$

p_i is the quark momentum, which we assume to be a fraction of the proton momentum: $p_i = z_i p$. In the lab frame with the proton at rest, the quarks must also be (roughly) at rest, so $p_i = m_i c$, $p = Mc$, and $m_i = z_i M$. This implies $x_i = x/z_i$. Plugging into the point-mass model gives

$$W_1^i = \frac{Q_i^2}{2M} \delta(x - z_i)$$

$$W_2^i = -\frac{2x^2 M c^2}{q^2} Q_i^2 \delta(x - z_i)$$

Let $f_i(z_i)$ be the probability of the i -th quark carrying a momentum fraction z_i . Then

$$W_1 = \sum_i \int_0^1 \frac{Q_i^2}{2M} \delta(x - z_i) f_i(z_i) dz_i = \frac{1}{2M} \sum_i Q_i^2 f_i(x)$$

$$W_2 = \sum_i \int_0^1 -\frac{2x^2 M c^2}{q^2} Q_i^2 \delta(x - z_i) f_i(z_i) dz_i = -\frac{2M c^2}{q^2} x^2 \sum_i Q_i^2 f_i(x)$$

The Bjorken scaling law: at high energies, the structure functions start to depend only on x , less so on q^2 .

$$M W_1 = \frac{1}{2} \sum_i Q_i^2 f_i(x) \longrightarrow F_1(x)$$

$$-\frac{q^2}{2M c^2 x} W_2 = x \sum_i Q_i^2 f_i(x) \longrightarrow F_2(x)$$

Callan-Gross relation:

$$F_2(x) = 2x F_1(x)$$

These two laws were experimentally confirmed and gave strength to the quark model.

8 Quantum Chromodynamics

8.1 Feynman Rules

8.1.1 Notation

QCD Uses Dirac spinors $u^{(s)}(p)$ for momentum and spin, but also the color vector c .

$$(\text{Red}) \ c = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad (\text{Blue}) \ c = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}; \quad (\text{Green}) \ c = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Gluons are massless and have spin 1. Represented by polarization vector ϵ^μ , orthogonal to momentum p^μ .

$$\text{Lorentz condition: } \epsilon^\mu p_\mu = 0$$

$$\text{Coulomb gauge: } \epsilon^0 = 0, \ \vec{\epsilon} \cdot \vec{p} = 0$$

Gluons have color (carry away one, give one). Mathematically, there should be 3×3 choices, but due to quark confinement, the color singlet $(r\bar{r} + b\bar{b} + g\bar{g})/\sqrt{3}$ isn't observed and we're left with 8.

$$\begin{aligned} |1\rangle &= (r\bar{b} + b\bar{r})/\sqrt{2} & |5\rangle &= -i(r\bar{g} - g\bar{r})/\sqrt{2} \\ |2\rangle &= -i(r\bar{b} - b\bar{r})/\sqrt{2} & |6\rangle &= (b\bar{g} - g\bar{b})/\sqrt{2} \\ |3\rangle &= (r\bar{r} - b\bar{b})/\sqrt{2} & |7\rangle &= -i(b\bar{g} - g\bar{b})/\sqrt{2} \\ |4\rangle &= (r\bar{g} + g\bar{r})/\sqrt{2} & |8\rangle &= (r\bar{r} + b\bar{b} - 2g\bar{g})/\sqrt{6} \end{aligned}$$

$a = (1\ 0\ 0\ 0\ 0\ 0\ 0\ 0)^T$ for $|1\rangle$, and $a = (0\ 0\ 0\ 0\ 0\ 0\ 1\ 0)^T$ for $|7\rangle$, etc.

We also have the Gell-Mann matrices that span $SU(3)$.

$$\begin{aligned}\lambda^1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} & \lambda^2 &= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} & \lambda^3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ \lambda^4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} & \lambda^5 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} & \lambda^6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \\ \lambda^7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} & \lambda^8 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}\end{aligned}$$

The commutator of the Gell-Mann matrices define the structure constants $f^{\alpha\beta\gamma}$ of $SU(3)$:

$$[\lambda^\alpha, \lambda^\beta] = 2if^{\alpha\beta\gamma}\lambda^\gamma$$

8.1.2 Procedure

1. For external quark lines with momentum p , spin s , and color c ,

(Quark) Incoming: $u^{(s)}(p)c$

(Quark) Outgoing: $\bar{u}^{(s)}(p)c^\dagger$

(Antiquark) Incoming: $\bar{\nu}^{(s)}(p)c^\dagger$

(Antiquark) Outgoing: $\nu^{(s)}(p)c$

For external gluon lines with momentum p , polarization ϵ , and color a ,

Incoming: $\epsilon_\mu(p)a^\alpha$

Outgoing: $\epsilon_\mu^*(p)a^{\alpha*}$

2. Each internal line contributes (going from (α, μ) to (β, ν))

Quark: $\frac{i(\not{q} + mc)}{q^2 - m^2c^2}$

Gluon: $\frac{-ig_{\mu\nu}\delta^{\alpha\beta}}{q^2}$

3. Each vertex contributes (here (α, μ) , (β, ν) , and (γ, λ) are incoming with gluons k_1, k_2, k_3 ; reverse signs of k if outgoing)

Quark-gluon: $\frac{-ig_s}{2}\lambda^\alpha\gamma^\mu$

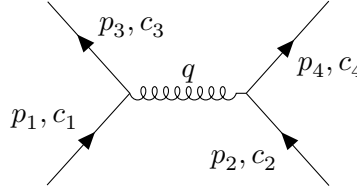
Three-gluon: $-g_sf^{\alpha\beta\gamma}[g_{\mu\nu}(k_1 - k_2)_\lambda + g_{\nu\lambda}(k_2 - k_3)_\mu + g_{\lambda\mu}(k_3 - k_1)_\nu]$

Four-gluon : $-ig_s^2[f^{\alpha\beta\eta}f^{\gamma\delta\eta}(g_{\mu\lambda}g_{\nu\rho} - g_{\mu\rho}g_{\nu\lambda}) + f^{\alpha\beta\eta}f^{\beta\gamma\eta}(g_{\mu\nu}g_{\lambda\rho} - g_{\mu\lambda}g_{\nu\rho}) + f^{\alpha\gamma\eta}f^{\delta\beta\eta}(g_{\mu\rho}g_{\nu\lambda} - g_{\mu\nu}g_{\lambda\rho})]$

4. Impose conservation of energy at each vertex with appropriate delta function, and erase final delta function to get $-i\mathcal{M}$.

8.2 Quark-Antiquark Interactions

Feynman diagram:



The amplitude after grouping terms is

$$\begin{aligned}\mathcal{M} &= -\frac{g_s^2}{q^2} \frac{1}{4} [\bar{u}(3)\gamma^\mu u(1)] [\bar{\nu}(2)\gamma_\mu \nu(4)] (c_3^\dagger \lambda^\alpha c_1) (c_2^\dagger \lambda^\alpha c_4) \\ \mathcal{M} &= -\frac{g_s^2}{q^2} [\bar{u}(3)\gamma^\mu u(1)] [\bar{\nu}(2)\gamma_\mu \nu(4)] f \\ f &= \frac{1}{4} (c_3^\dagger \lambda^\alpha c_1) (c_2^\dagger \lambda^\alpha c_4)\end{aligned}$$

$\mathcal{M}$ has a similar structure to that of electron-positron scattering, with g_s instead of g_e . There's also a **color factor** f . Thus the potential in QCD can be thought of as

$$V_{q\bar{q}}(r) = -f \frac{\alpha_s \hbar c}{r}$$

- For octets, e.g. $r\bar{b}$, $f = -1/6$.
- For color singlets, $(r\bar{r} + b\bar{b} + g\bar{g})/\sqrt{3}$, $f = 4/3$.

Thus $V_{q\bar{q}}(r)$ is attractive for octets and repulsive for singlets.

8.3 Quark-Quark Interactions

Quark-quark interactions are very similar to quark-antiquark interactions. However, now a color octet/singlet isn't possible. Instead, we have:

$$\begin{aligned}\text{Triplet} & \begin{cases} (rb - br)/\sqrt{2} \\ (bg - gb)/\sqrt{2} \\ (gr - rg)/\sqrt{2} \end{cases} \\ \text{Sextet} & \begin{cases} rr, bb, gg \\ (rb + br)/\sqrt{2}, (bg + gb)/\sqrt{2}, (rg + gr)/\sqrt{2} \end{cases}\end{aligned}$$

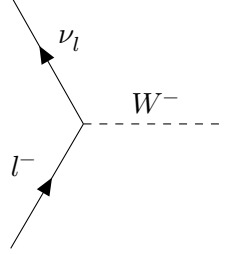
- For triplets: $f = -2/3$
- For sextets: $f = 1/3$

Thus $V_{qq}(r)$ is attractive for triplets and repulsive for sextets.

9 Weak Interactions

9.1 Charged Weak Interactions of Leptons

The fundamental leptonic vertex:



The propagator for $W^\pm, Z^0$ is

$$\frac{-i(g_{\mu\nu} - q_\mu q_\nu / M^2 c^2)}{q^2 - M^2 c^2} \rightarrow \frac{ig_{\mu\nu}}{(Mc)^2} \text{ at } q^2 \ll (Mc)^2$$

The weak vertex factor is

$$\frac{-ig_w}{2\sqrt{2}} \gamma^\mu (1 - \gamma^5), \quad g_w = \sqrt{4\pi\alpha_w}$$

Otherwise, the Feynman rules are the same. The weak interaction is overshadowed by the stronger electromagnetic interaction, so evidence for it is often implicit.

These Feynman rules can be used to calculate the weak decay of muons, neutrons, pions, etc. (See Griffiths 10.2, 10.3, 10.4)

Note: conservation of electron number, muon number, and tau number dictate that $W^\pm$ couple only between particles of a single generation:

$$\begin{pmatrix} \nu_e \\ e \end{pmatrix}, \quad \begin{pmatrix} \nu_\mu \\ \mu \end{pmatrix}, \quad \begin{pmatrix} \nu_\tau \\ \tau \end{pmatrix}$$

9.2 Charged Weak Interaction of Quarks

Unlike leptonic interactions, weak interactions of quarks aren't restricted to quarks of the same generation. Instead, the weak interaction quark generations are

$$\begin{pmatrix} u \\ d' \end{pmatrix}, \quad \begin{pmatrix} c \\ s' \end{pmatrix}, \quad \begin{pmatrix} t \\ b' \end{pmatrix}$$

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \begin{pmatrix} U_{ud} & U_{us} & U_{ub} \\ U_{cd} & U_{cs} & U_{cb} \\ U_{td} & U_{ts} & U_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

This Kobayashi-Maskawa matrix is given by (with $c_i = \cos \theta_i$ and $s_i = \sin \theta_i$)

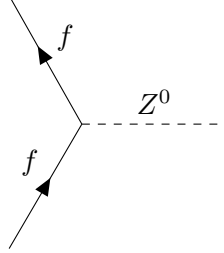
$$U = \begin{pmatrix} c_1 & s_1 c_3 & s_1 s_3 \\ -s_1 c_2 & c_1 c_2 c_3 - s_2 s_3 e^{i\delta} & c_1 c_2 s_3 + s_2 c_3 e^{i\delta} \\ -s_1 s_2 & c_1 s_2 c_3 + c_2 s_3 e^{i\delta} & c_1 s_2 s_3 - c_2 c_3 e^{i\delta} \end{pmatrix}$$

The appropriate factors are then multiplied to the vertex factors:

$$\frac{-ig_w}{2\sqrt{2}}\gamma^\mu(1-\gamma^5) \times (\text{KM matrix factors})$$

9.3 Neutral Weak Interactions

The fundamental vertex:



The vertex factor of Z^0 is

$$\frac{-ig_z}{2}\gamma^\mu(c_V^f - c_A^f\gamma^5)$$

f	c_V	c_A
ν_e, ν_μ, ν_τ	$\frac{1}{2}$	$\frac{1}{2}$
e^-, μ^-, τ^-	$-\frac{1}{2} + 2\sin\theta_w$	$-\frac{1}{2}$
u, c, t	$\frac{1}{2} - \frac{4}{3}\sin^2\theta_w$	$\frac{1}{2}$
d, s, b	$-\frac{1}{2} + \frac{2}{3}\sin^2\theta_w$	$-\frac{1}{2}$

The propagator of Z^0 is

$$\frac{-i(g_{\mu\nu} - q_\mu q_\nu / M_Z^2 c^2)}{q^2 - M_Z^2 c^2} \rightarrow \frac{ig_{\mu\nu}}{(M_Z c)^2} \text{ when } q^2 \ll (M_Z c)^2$$

To account for the finite lifetime of Z^0 , say τ_z with decay rate $\Gamma_z = 1/\tau_z$, we can use

$$\frac{-i(g_{\mu\nu} - q_\mu q_\nu / M_Z^2 c^2)}{q^2 - M_Z^2 c^2 + i\hbar M_Z \Gamma_Z}$$

9.3.1 The GWS Model

Above, θ_w is the weak mixing angle. Its value is given by experiment: $\theta_w \approx 28.7^\circ$. It defines the relationship between the weak and electromagnetic coupling constants.

$$g_w = \frac{g_e}{\sin\theta_w}, \quad g_z = \frac{g_e}{\sin\theta_w \cos\theta_w}$$

It also relates the $W^\pm$ and Z^0 masses:

$$M_W = M_Z \cos\theta_w$$

The theoretical derivation of θ_w hasn't happened yet in the Standard Model...

10 Gauge Theories

10.1 A Family of Lagrangians

In relativistic field theory, the Lagrangian formulation is similar to classical:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i} \quad \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_i)} \right) = \frac{\partial \mathcal{L}}{\partial \phi_i}$$

Lagrangians are constructed as axioms. Usually no justification other than “they give the right equations of motion.”

10.1.1 Klein-Gordon: Scalar Field, Spin 0

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \phi) (\partial^\mu \phi) - \frac{1}{2} \left(\frac{mc}{\hbar} \right)^2 \phi^2$$

Euler-Lagrange equation gives the Klein-Gordon equation for a scalar field of spin 0.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} &= \partial^\mu \phi, \quad \frac{\partial \mathcal{L}}{\partial \phi} = - \left(\frac{mc}{\hbar} \right)^2 \phi \\ \partial_\mu \partial^\mu \phi + \left(\frac{mc}{\hbar} \right)^2 \phi &= 0 \end{aligned}$$

10.1.2 Dirac: Spinor Field, Spin $\frac{1}{2}$

$$\mathcal{L} = i(\hbar c) \bar{\psi} \gamma^\mu \partial_\mu \psi - (mc^2) \bar{\psi} \psi$$

Euler-Lagrange for $\bar{\psi}$ gives

$$i\gamma^\mu \partial_\mu \psi - \frac{mc}{\hbar} \psi = 0$$

Euler-Lagrange for ψ gives

$$i\partial_\mu \bar{\psi} \gamma^\mu + \frac{mc}{\hbar} \bar{\psi} = 0$$

10.1.3 Proca: Vector Field, Spin 1

$$\mathcal{L} = -\frac{1}{16\pi} F^{\mu\nu} F_{\mu\nu} + \frac{1}{8\pi} \left(\frac{mc}{\hbar} \right)^2 A^\nu A_\nu$$

Here $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$. Euler-Lagrange equation gives

$$\partial_\mu F^{\mu\nu} + \left(\frac{mc}{\hbar} \right)^2 A^\nu = 0$$

10.1.4 Maxwell's Equations

$$\mathcal{L} = -\frac{1}{16\pi} F^{\mu\nu} F_{\mu\nu} - \frac{1}{c} J^\mu A_\mu$$

Euler-Lagrange equation gives

$$\partial_\mu F^{\mu\nu} = \frac{4\pi}{c} J^\nu$$

10.2 Local $U(1)$ Gauge Invariance and QED

The Dirac Lagrangian $\mathcal{L} = i\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi$ is invariant under global gauge transformations $\psi \rightarrow e^{i\theta} \psi$. However it isn't under the local gauge transformations $\psi \rightarrow U(1)\psi$, where $U(1)$ is a 1×1 unitary matrix $UU^\dagger = I$.

Take $U = e^{i\theta(x)}$, so $\psi \rightarrow e^{i\theta(x)} \psi$. Then

$$\begin{aligned} \mathcal{L} &\rightarrow \mathcal{L} - \hbar c (\partial_\mu \theta) \bar{\psi} \gamma^\mu \psi = \mathcal{L} + (q \bar{\psi} \gamma^\mu \psi) \partial_\mu \lambda \\ \lambda(x) &= -\frac{\hbar c}{q} \theta(x) \end{aligned}$$

We demand that the total Lagrangian is invariant under these $U(1)$ local gauge transformations. Thus we introduce an extra term A_μ that transforms as $A_\mu \rightarrow A_\mu + \partial_\mu \lambda$.

$$\begin{aligned} \mathcal{L} &= [i\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi] - (q \bar{\psi} \gamma^\mu \psi) A_\mu \\ \psi &\rightarrow e^{-iq\lambda(x)/\hbar c} \psi \\ A_\mu &\rightarrow A_\mu + \partial_\mu \lambda \end{aligned}$$

Since A^μ is a vector field, it obeys the Proca Lagrangian. The cross term $A^\nu A_\nu$ in Proca is not invariant under $A_\mu \rightarrow A_\mu + \partial_\mu \lambda$, thus we require $A^\nu A_\nu = m^2 c^2 = 0$, or $m_A = 0$. The introduced field must be massless.

In total, the Dirac Lagrangian under local $U(1)$ gauge invariance becomes

$$\mathcal{L} = [i\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi] - \frac{1}{16\pi} F^{\mu\nu} F_{\mu\nu} - [(q \bar{\psi} \gamma^\mu \psi) A_\mu]$$

This is the QED Lagrangian describing the interaction of Dirac fields (electrons, protons) with Maxwell fields (massless photons).

Note that the transformation is

$$\partial_\mu \psi \rightarrow e^{-iq\lambda(x)/\hbar c} \left[\partial_\mu - i \frac{q}{\hbar c} (\partial_\mu \lambda) \right] \psi$$

By replacing every ∂_μ with covariant derivatives (a la general relativity) $\mathcal{D}_\mu$, we can enforce gauge invariance automatically.

$$\begin{aligned} \mathcal{D}_\mu &= \partial_\mu + i \frac{q}{\hbar c} A_\mu \\ \mathcal{D}_\mu \psi &\rightarrow e^{-iq\lambda(x)/\hbar c} \mathcal{D}_\mu \psi \end{aligned}$$

To recap:

- We enforced local $U(1)$ gauge invariance.
- We found that this required a massless vector field A^μ .
- We interpreted this as the QED Lagrangian describing interactions of Dirac fields with Maxwell fields.

10.3 Local $SU(2)$ Gauge Invariance and Yang-Mills Theory

We consider two spin- $\frac{1}{2}$ fields ψ_1, ψ_2 in the spinor $\psi = (\psi_1 \ \psi_2)$. If the masses are equal, the total Dirac Lagrangian is

$$\mathcal{L} = i\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi$$

We require the global $SU(2)$ gauge invariance to hold locally: $\psi \rightarrow S\psi = e^{i\theta} e^{i\tau \cdot \mathbf{a}} \psi$.

- General 2×2 Hermitian matrix has form $H = \theta I + \tau \cdot \mathbf{a}$

$$\tau = \begin{pmatrix} \sigma_x & \sigma_y & \sigma_z \end{pmatrix} \quad \text{(Pauli matrices)}$$

$$\mathbf{a} = \begin{pmatrix} a_1 & a_2 & a_3 \end{pmatrix} \quad \text{(Real numbers)}$$

- General 2×2 unitary matrix has form $U = e^{iH} = e^{i\theta} e^{i\tau \cdot \mathbf{a}}$

Let $\mathbf{a}$ depend on x and define $\lambda(x) = -(\hbar c/q)\mathbf{a}(x)$. Enforce local $SU(2)$ gauge invariance. As before, define the covariant derivative $\mathcal{D}_\mu$ such that

$$\begin{aligned} \mathcal{D}_\mu &= \partial_\mu + i \frac{q}{\hbar c} \tau \cdot \mathbf{A}_\mu \\ \mathbf{A}_\mu &\rightarrow \mathbf{A}_\mu + \partial_\mu \lambda + \frac{2q}{\hbar c} (\lambda \times \mathbf{A}_\mu) \end{aligned}$$

This leads to the modified Lagrangian:

$$\mathcal{L} = [i\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi] - (q \bar{\psi} \gamma^\mu \tau \psi) \cdot \mathbf{A}_\mu$$

$\mathbf{A}_\mu$ are the 3 gauge fields that transform according to above. This gauge field, as a spin-1 massless vector field (see prev. section's arguments), has its own Lagrangian

$$\begin{aligned} \mathcal{L}_A &= -\frac{1}{16\pi} \mathbf{F}^{\mu\nu} \cdot \mathbf{F}_{\mu\nu} \\ \mathbf{F}^{\mu\nu} &\equiv \partial^\mu \mathbf{A}^\nu - \partial^\nu \mathbf{A}^\mu - \frac{2q}{\hbar c} (\mathbf{A}^\mu \times \mathbf{A}^\nu) \end{aligned}$$

The complete Yang-Mills Lagrangian for 2 spin- $\frac{1}{2}$ Dirac fields is

$$\mathcal{L} = [i\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi] - \frac{1}{16\pi} \mathbf{F}^{\mu\nu} \cdot \mathbf{F}_{\mu\nu} - (q \bar{\psi} \gamma^\mu \tau \psi) \cdot \mathbf{A}_\mu$$

With $\mathbf{F}^{\mu\nu}$ defined above. To recap:

- We started from the 2-particle Lagrangian.
- We enforced local $SU(2)$ gauge invariance.
- We got the Yang-Mills Lagrangian.

10.4 Local $SU(3)$ Gauge Invariance and QCD

Can be seen as an application of Yang-Mills. We have 3 spin- $\frac{1}{2}$ fields now, for 3 color quarks of equal mass. The Dirac Lagrangian:

$$\mathcal{L} = i\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi$$

We enforce local $SU(3)$ gauge invariance: $\psi \rightarrow S\psi = e^{i\theta} e^{i\lambda \cdot \mathbf{a}(x)} \psi$

- General 3×3 Hermitian matrix has form $H = \theta I + \lambda \cdot \mathbf{a}$

$$\lambda = 8 \text{ Gell-Mann matrices}$$

$$\mathbf{a} = 8 \text{ real numbers}$$

- Any 3×3 unitary matrix has form $U = e^{iH} = e^{i\theta} e^{i\lambda \cdot \mathbf{a}}$

Define $\phi(x) = -(\hbar c/q)\mathbf{a}(x)$ and find the covariant derivative to be

$$\begin{aligned} \mathcal{D}_\mu &= \partial_\mu + i \frac{q}{\hbar c} \lambda \cdot \mathbf{A}_\mu \\ \mathbf{A}_\mu &\rightarrow \mathbf{A}_\mu + \partial_\mu \phi + \frac{2q}{\hbar c} (\phi \times \mathbf{A}_\mu) \\ SU(3) \text{ cross product: } (\mathbf{B} \times \mathbf{C})_i &= \sum_{j,k=1}^8 f_{ijk} B_j C_k \end{aligned}$$

The modified Lagrangian ($\mathcal{D}$ instead of ∂_μ) is:

$$L = [i\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi] - (q \bar{\psi} \gamma^\mu \lambda \psi) \cdot \mathbf{A}_\mu$$

We also need to add the free gluon Lagrangians (8 of them, one for each λ_i):

$$\begin{aligned} \mathcal{L} &= -\frac{1}{16\pi} \mathbf{F}^{\mu\nu} \cdot \mathbf{F}_{\mu\nu} \\ \mathbf{F}^{\mu\nu} &\equiv \partial^\mu \mathbf{A}^\nu - \partial^\nu \mathbf{A}^\mu - \frac{2q}{\hbar c} (\mathbf{A}^\mu \times \mathbf{A}^\nu) \end{aligned}$$

The total chromodynamics Lagrangian is:

$$\mathcal{L} = [i\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi] - \frac{1}{16\pi} \mathbf{F}^{\mu\nu} \cdot \mathbf{F}_{\mu\nu} - (q \bar{\psi} \gamma^\mu \lambda \psi) \cdot \mathbf{A}_\mu$$

With $\mathbf{F}^{\mu\nu}$ and the $SU(3)$ cross product defined above. To recap:

- We enforced local $SU(3)$ gauge invariance in the 3-particle Dirac Lagrangian.
- We had to introduce 8 gauge fields.
- This was interpreted as 8 gluons for 3 color quarks.
- This Lagrangian is thought to describe the strong interaction, a measurable application of Yang-Mills theory.

10.5 The Higgs Mechanism

10.5.1 Interpretation of Mass in Lagrangian

1. Find a ground state for which $V = \min$ in the Lagrangian. E.g. $\phi = \eta$
2. Re-express Lagrangian in terms of η .
3. Mass term is coefficient of η^2 .

E.g. Assume we have a scalar field and Lagrangian

$$\mathcal{L} = \underbrace{\frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi)}_T + \underbrace{\frac{1}{2}\mu^2 \phi^2 - \frac{1}{4}\lambda^2 \phi^4}_V$$

Klein-Gordon Lagrangian for scalar fields predicts mass coefficient $-\frac{1}{2}(mc/\hbar)^2$. A naive interpretation of $\frac{1}{2}\mu^2$ (coefficient of ϕ^2) as the mass coefficient gives $m = \pm i\mu\hbar/c$, and an imaginary mass isn't possible.

Instead, note that V has its minimum at $\phi = \pm\mu/\lambda$. Rewrite the Lagrangian in terms of $\eta = \phi \pm \mu/\lambda$.

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) + \frac{1}{2}\mu^2 \left(\eta \pm \frac{\mu}{\lambda}\right)^2 - \frac{1}{4}\lambda^2 \left(\eta \pm \frac{\mu}{\lambda}\right)^4$$

Coefficient of η^2 is $-\mu^2$. Setting equal to the Klein-Gordon mass coefficient, we get $m = \sqrt{2}\mu\hbar/c$, much more reasonable.

10.5.2 Spontaneous Symmetry Breaking

In the previous example, $\mathcal{L}$ was invariant under $\phi \rightarrow -\phi$. Now it isn't. This is because the 2 ground states (vacuum states) don't agree.

Spontaneous symmetry breaking: symmetry is hidden by selection of 1 vacuum state over other.

Goldstone's theorem: Spontaneous breaking of continuous symmetry is always accompanied by a massless scalar particle, the **Goldstone boson**. E.g. for the 2-scalar field Lagrangian,

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi_1)(\partial^\mu \phi_1) + \frac{1}{2}(\partial_\mu \phi_2)(\partial^\mu \phi_2) + \frac{1}{2}\mu^2(\phi_1^2 + \phi_2^2) - \frac{1}{4}\lambda^2(\phi_1^2 + \phi_2^2)^2$$

The minima lie on $\phi_1^2 + \phi_2^2 = \mu^2/\lambda^2$. Might as well pick $\phi_1 = \mu/\lambda$, $\phi_2 = 0$. Let $\eta = \phi_1 - \mu/\lambda$ and $\xi = \phi_2$. Then the Lagrangian becomes

$$\begin{aligned} \mathcal{L} = & \left[\frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) - \mu^2 \eta^2 \right] + \left[\frac{1}{2}(\partial_\mu \xi)(\partial^\mu \xi) \right] \\ & + \left[\mu\lambda(\eta^3 + \eta\xi^2) - \frac{\lambda^2}{4}(\eta^4 + \xi^4 + 2\eta^2\xi^2) \right] + \frac{\mu^4}{4\lambda^2} \end{aligned}$$

Interpret first two terms as the Klein-Gordon Lagrangians, and get $m_\eta = \sqrt{2}\mu\hbar/c$ and $m_\xi = 0$. The latter is the massless Goldstone boson. Remaining terms are couplings and an ignorable scalar term.

10.5.3 The Higgs Mechanism

To simplify the previous example, let $\phi \equiv \phi_1 + i\phi_2$. The Lagrangian becomes

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^*(\partial^\mu \phi) + \frac{1}{2}\mu^2(\phi^*\phi) - \frac{1}{4}\lambda^2(\phi^*\phi)^2$$

Previous minima were on $\phi_1^2 + \phi_2^2 = \mu^2/\lambda^2$, a $SO(2)$ invariance. In terms of ϕ , this is global $U(1)$ invariance under $\phi \rightarrow e^{i\theta}\phi$. We enforce *local* $U(1)$ invariance and replace ∂_μ with $\mathcal{D}_\mu$.

$$\begin{aligned} \mathcal{D}_\mu &= \partial_\mu + i\frac{q}{\hbar c}A_\mu, \quad A_\mu \text{ is massless vector field} \\ \mathcal{L} &= \frac{1}{2}(\mathcal{D}_\mu \phi)^*(\mathcal{D}^\mu \phi) + \frac{1}{2}\mu^2(\phi^*\phi) - \frac{1}{4}\lambda^2(\phi^*\phi)^2 - \frac{1}{16\pi}F^{\mu\nu}F_{\mu\nu} \end{aligned}$$

Retracing our steps, set $\eta = \phi_1 - \mu/\lambda$ and $\xi = \phi_2$. Then

$$\begin{aligned} \mathcal{L} &= \left[\frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) - \mu^2\eta^2 \right] + \left[\frac{1}{2}(\partial_\mu \xi)(\partial^\mu \xi) \right] \\ &\quad + \left[-\frac{1}{16\pi}F^{\mu\nu}F_{\mu\nu} + \frac{1}{2}\left(\frac{q\mu}{\hbar c\lambda}\right)^2 A_\mu A^\mu \right] - 2i\left(\frac{q\mu}{\hbar c\lambda}\right)(\partial_\mu \xi)A^\mu + \dots \end{aligned}$$

The first two terms describe Klein-Gordon scalar fields and a massless Goldstone boson. The third term is the Lagrangian for A^μ , but now it has mass: $m = (\sqrt{4\pi}q\mu)/(\lambda c^2)$ after comparing with the Proca Lagrangian.

To get rid of the imaginary term, exploit gauge invariance and choose θ such that $\phi_2 = \xi = 0$. E.g. $\theta = -\tan^{-1}(\phi_2/\phi_1)$.

$$\mathcal{L} = \left[\frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) - \mu^2\eta^2 \right] + \left[-\frac{1}{16\pi}F^{\mu\nu}F_{\mu\nu} + \frac{1}{2}\left(\frac{q\mu}{\hbar c\lambda}\right)^2 A_\mu A^\mu \right] + \dots$$

The Goldstone boson has been eliminated. We have a single massive scalar η left, this is the **Higgs particle**.

To recap, the Higgs mechanism:

1. Spontaneous symmetry breaking in 2-scalar particle Lagrangian
2. Gauge field absorbs Goldstone boson to acquire mass
3. This is responsibly for the mass of $W^\pm$ and Z^0 .

*Note: This is just an example, and the Lagrangian probably isn't of this exact form.